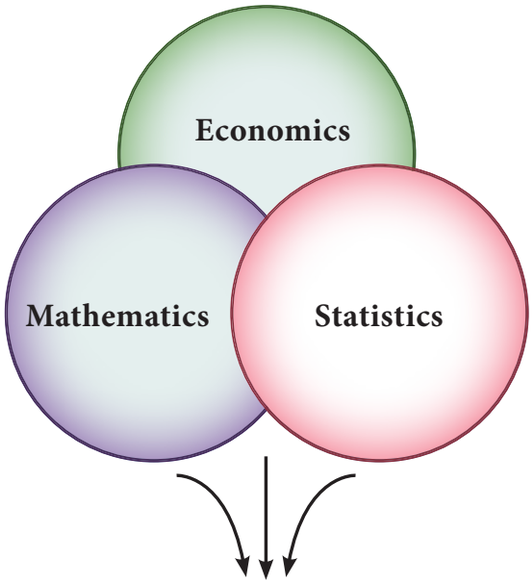


Amalgamation of above Three Subjects is Econometrics

Methodology Of Econometrics

Broadly speaking, traditional or classical econometric methodology consists of the following steps.

1) Statement of the theory or hypothesis
2) Specification of the mathematical model of the theory
3) Specification of the econometric model of the theory
4) Obtaining the data
5) Estimation of the parameters of the econometric model
6) Hypothesis testing
7) Forecasting or prediction
8) Using the model for control or policy purposes.

Flow Chart of Anatomy / Methodology of Econometrics

Anatomy of Econometric Modeling

Economic theory

Mathematical model of the theory

Economic model of the theory

Data

Estimation of econometric model

Hypothesis Testing

Forecasting or prediction

Using the model for control or policy purpose

Difference between the Econometric model with Mathematical models and statistical models

1. Models in Mathematical Economics are developed based on Economic Theories, while, Econometric Models are developed based on Economic Theories to test the validity of Economic Theories in reality through the actual data.
2. Regression Analysis in Statistics does not concentrate more on error term while Econometric Models concentrate more on error terms

Statistics Regression:

$$Yi = \beta_0 + \beta_1 X_i$$

Econometrics Regression:

$$Yi = \beta_0 + \beta_1 X_i + U_i$$

(with more than 2 variables) or

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + U_i$$